Name

Mathematics Stage 5 core

Graphing Lines

Book 2 Recognise and graph equations
Examine parallel, horizontal and vertical lines

Version: 261006

Feedback:
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Syllabus Content

MA4-LIN-C-01 determines the midpoint, gradient and length of an interval, and graphs linear relationships, with and without digital tools

Recognise and graph equations

- Recognise that equations of the form $y = mx + c$ represent linear relationships or straight lines
- Construct tables of values and use coordinates to graph a variety of linear relationships on the Cartesian plane, with and without digital tools
- Identify the x- and y- intercepts of lines
- Determine whether a point lies on a line using substitution

Examine parallel, horizontal and vertical lines

- Explain that parallel lines have equal gradients/slopes
- Explain why the x-axis has the equation $y = 0$ and the y-axis has the equation $x = 0$
- Recognise $y = c$ as a line parallel to the x-axis and $x = k$ as a line parallel to the y-axis
- Graph vertical and horizontal lines

Sketching Lines using a Table

Review

☐ Substitute into equations

a Substitute $x = 3$ into $2x + 5$

b Substitute $x = 3$ into $5 - 2x$

c Substitute $x = -3$ into $2x + 5$

☐ Complete a table of values

a $y = x + 2$

x	0	1	2	3	4
y					
	$0+2$	$1+2$	$2+2$	$3+2$	$4+2$

b $y = 3x + 6$

x	0	1	2	3	4
y					
	$3\times0+6$	$3\times1+6$	$3\times2+6$	$3\times3+6$	$3\times4+6$

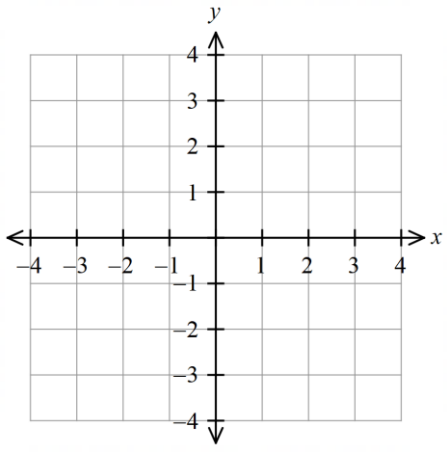
c $y = 8 - 2x$

x	0	1	2	3	4
y					
	$8-2\times0$	$8-2\times1$	$8-2\times2$	$8-2\times3$	$8-2\times4$

☐ Identify coordinates from a table of values

Plot this table of values on the Cartesian plane to the right.

x	-1	0	1
y	-1	1	3



ⓘ Equation of a Linear Relationship

Every linear relationship is in the form:

$$y = mx + c$$

ⓘ Identify equation of a linear relationship

✗	✓	✓	✓	✗
$y = x^2$	$y = 2x$	$y = 9 - 12x$	$y = \frac{x}{6} - 22$	$y = \frac{6}{x} - 22$

Identify which of the following are linear relationships

a $y = 5x + 1$	**b** $y = 1 + 5x$	**c** $y = x^2 - 7$	**d** $y = \frac{x}{2} - 7$
e $y = \frac{2}{x}$	**f** $y = 3^x$	**g** $y = x$	**h** $y = 9$
i $y = 3 - 7x$	**j** $y = -12x + 6$	**k** $y = x^3 - 5$	**l** $y = 80 - \frac{5x}{3}$

Sketch a Line from a Table of Values

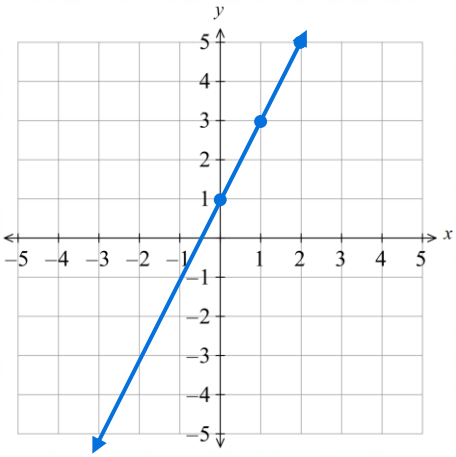
Some equations are called **linear relationships** because when we plot their x-y pairs, it forms a straight line.

1. Construct a table of values by substituting x values into the equation to find y.
2. Plot the (x, y) points on a Cartesian plane.
3. Draw a straight line through the points and extend it.

Example Graph a linear relationship using a table of values

Draw the graph of the equation $y = 2x + 1$

x	0	1	2	3
y	1	3	5	7

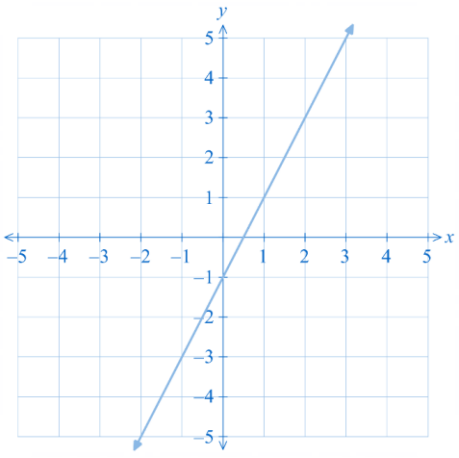


Guided Practice

Draw the graph of:

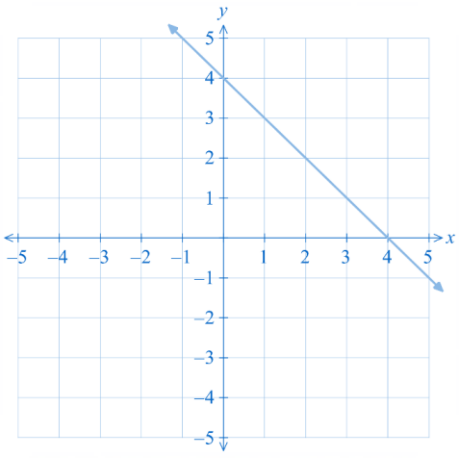
a $y = 2x - 1$

x	0	1	2	3
y				



b $y = 4 - x$

x	0	1	2	3
y				



Key Ideas

1. Each column of a table of values can be written as a ……
2. When an equation forms points that make a straight line, we call it a …… relationship.

1 Identify the coordinates from the table of values.

a

x	1	2	3	4
y	2	5	8	11
Coordinate	(1,2)	(2,5)	(3,8)	(4,11)

b

x	0	1	2	3
y	1	1.5	2	2.5
Coordinate	(0,1)	(1,1.5)	(2,2)	(3,2.5)

c

x	−1	0	1	2
y	7	5	3	1
Coordinate	(−1,7)	(0,5)	(1,3)	(2,1)

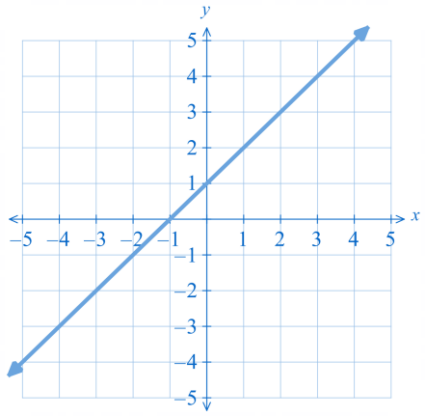
d

x	−2	−1	0	1
y	14	10	6	2
Coordinate	(−2,14)	(−1,10)	(0,6)	(1,2)

2 Complete the table of values and graph the result.

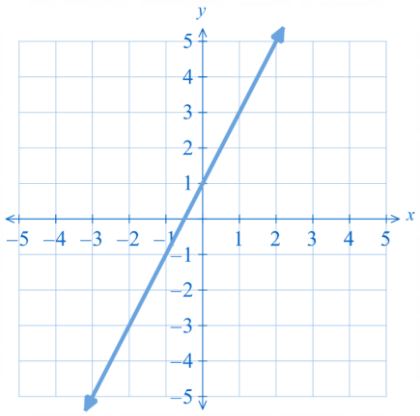
a $y = x + 1$

x	0	1	2	3
y				



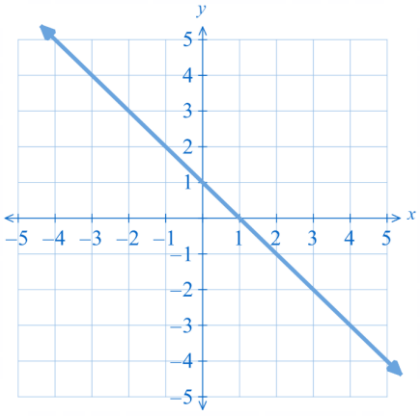
b $y = 2x + 1$

x	0	1	2	3
y				



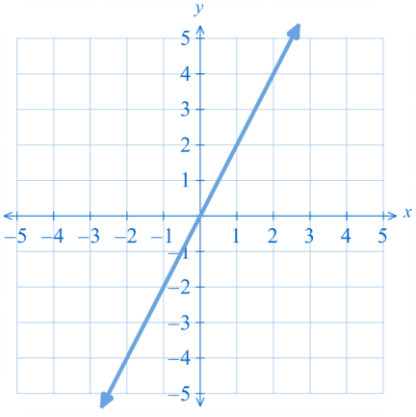
c $y = -x + 1$

x	0	1	2	3
y				



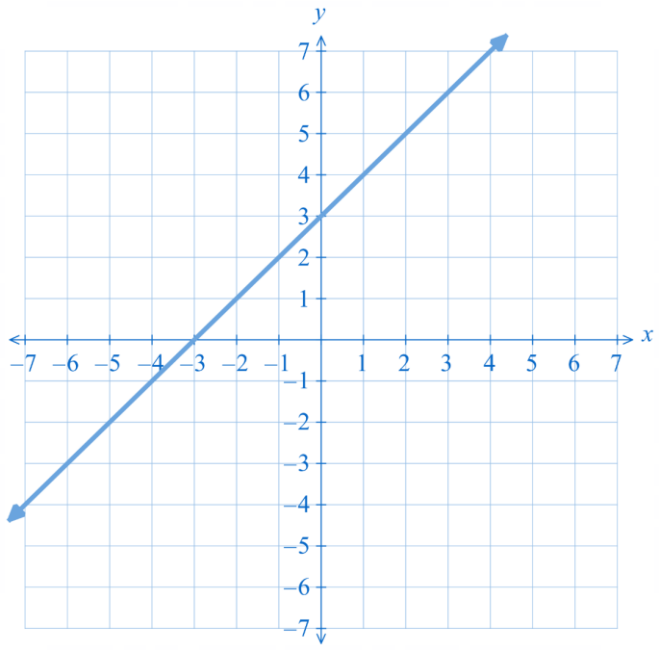
d $y = 2x$

x	0	1	2	3
y				



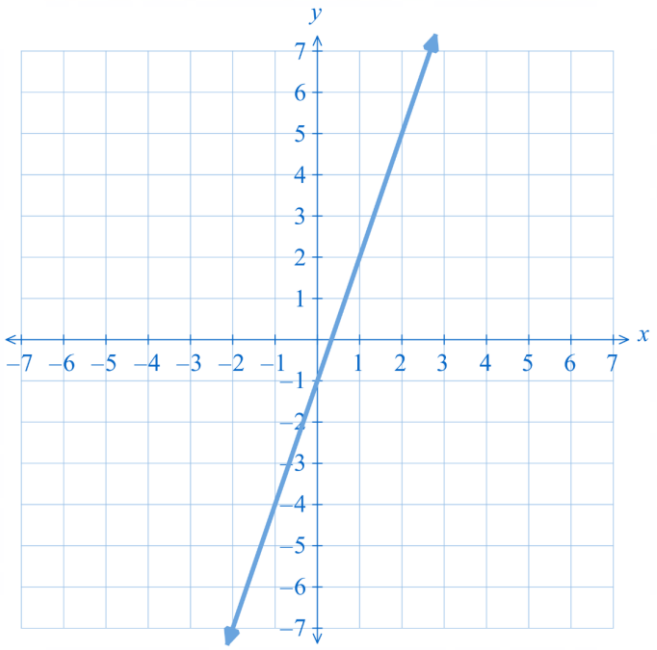
e $y = x + 3$

x	0	1	2	3
y				



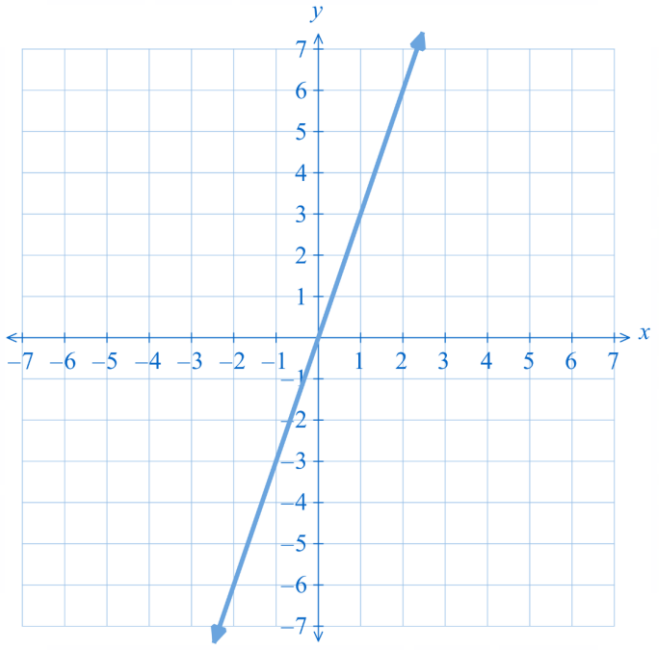
f $y = 3x - 1$

x	0	1	2	3
y				



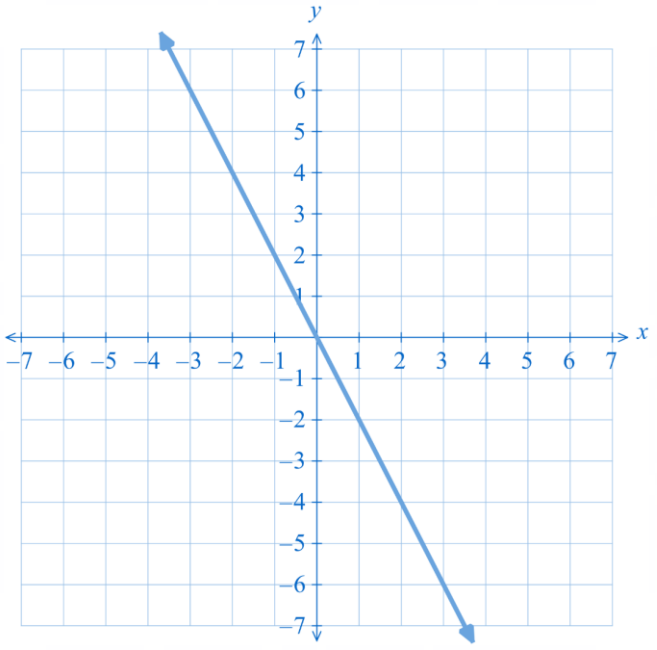
g $y = 3x$

x	0	1	2	3
y				



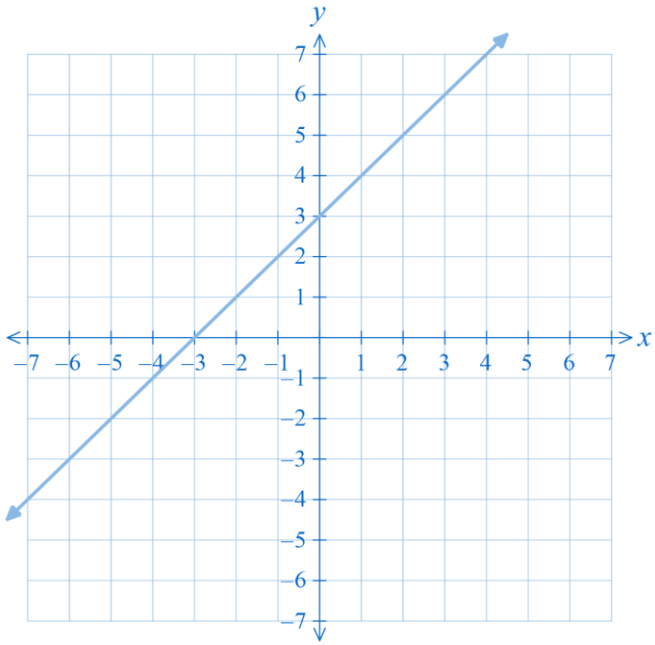
h $y = -2x$

x	0	1	2	3
y				



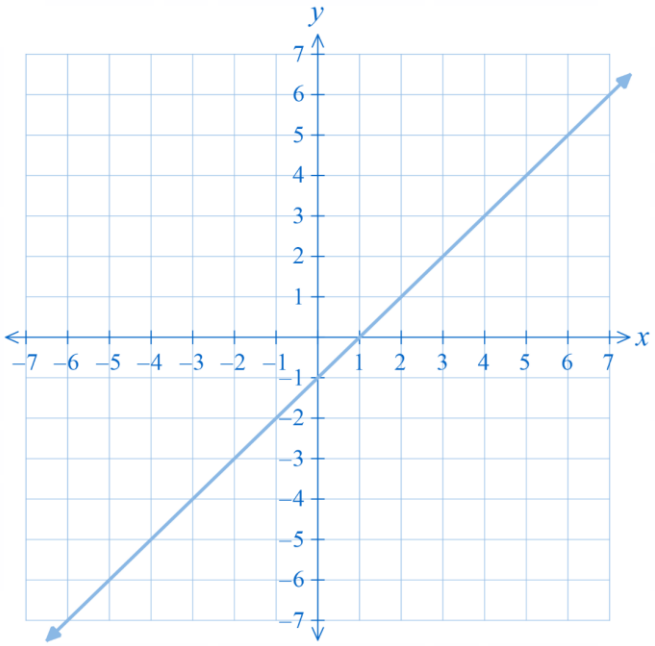
i $y = x + 3$

x	-2	-1	0	1	2
y					



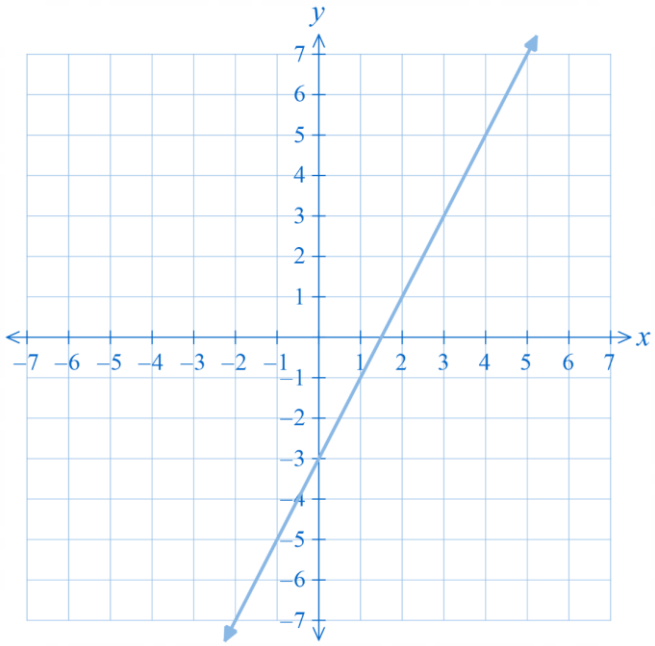
j $y = x - 1$

x	-2	-1	0	1	2
y					



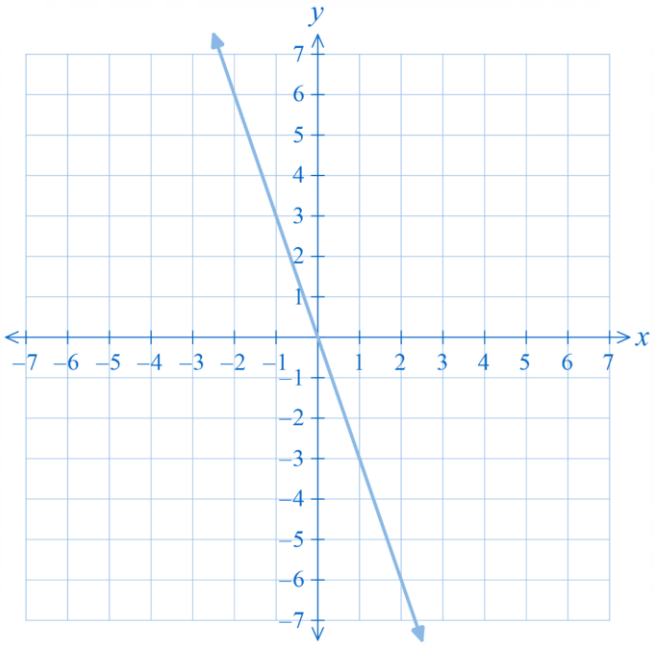
k $y = 2x - 3$

x	-2	-1	0	1	2
y					



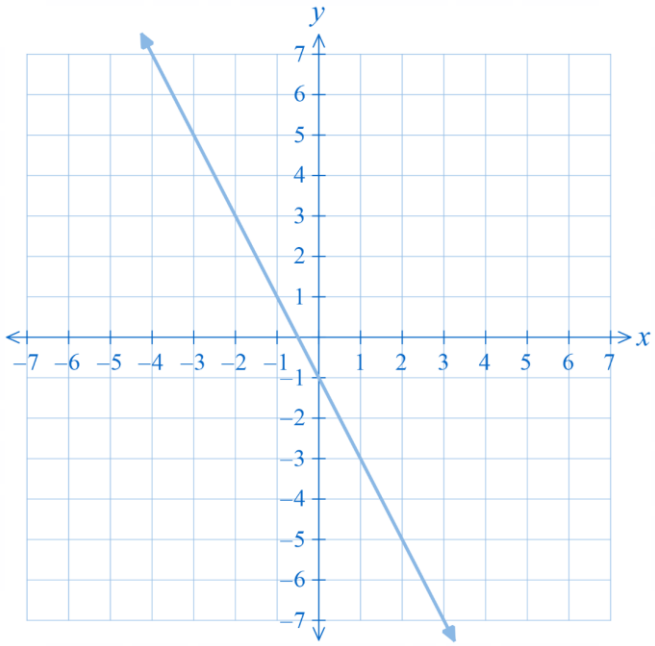
l $y = -3x$

x	-2	-1	0	1	2
y					



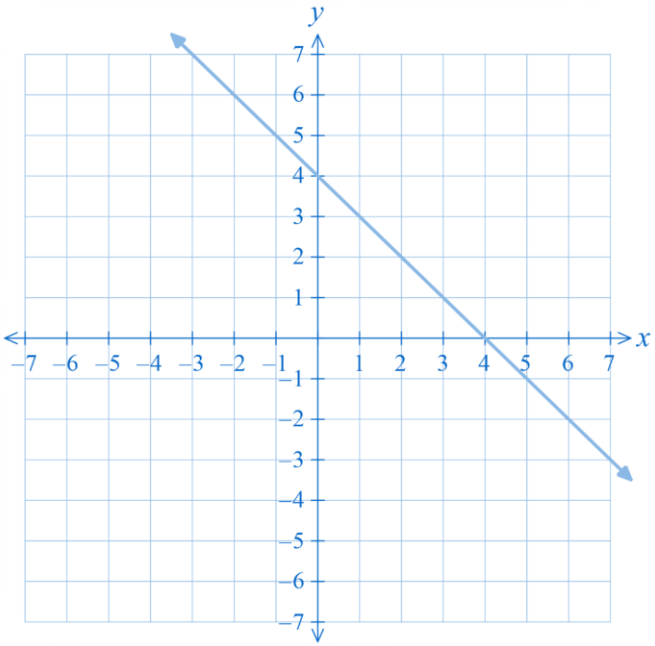
m $y = -2x - 1$

x	-2	-1	0	1	2
y					



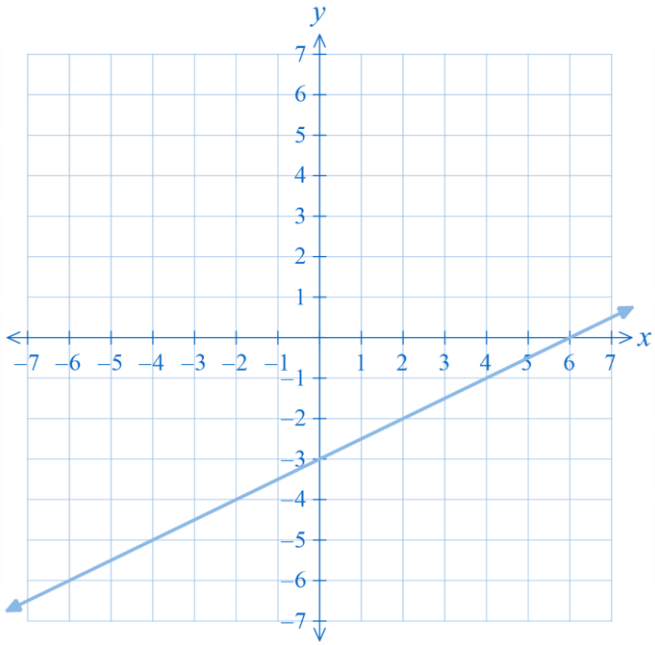
n $y = -x + 4$

x	-2	-1	0	1	2
y					



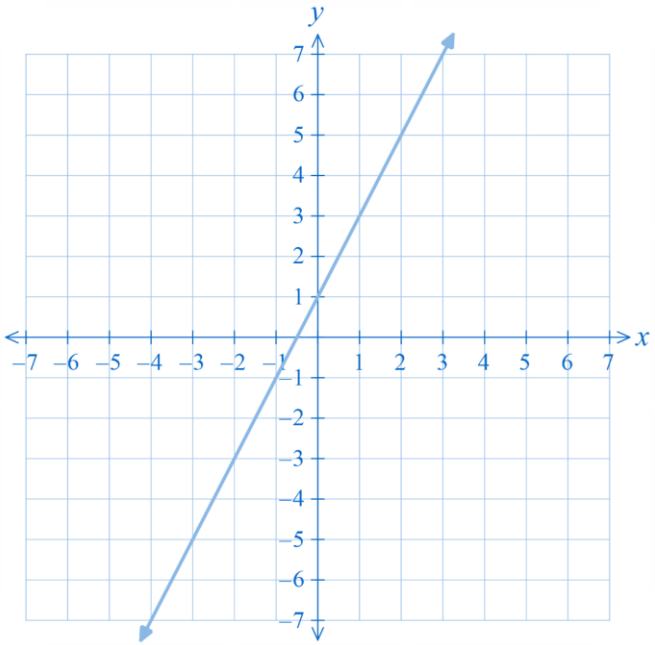
o $y = 0.5x - 3$

x	-2	-1	0	1	2
y					



p $y = 1 + 2x$

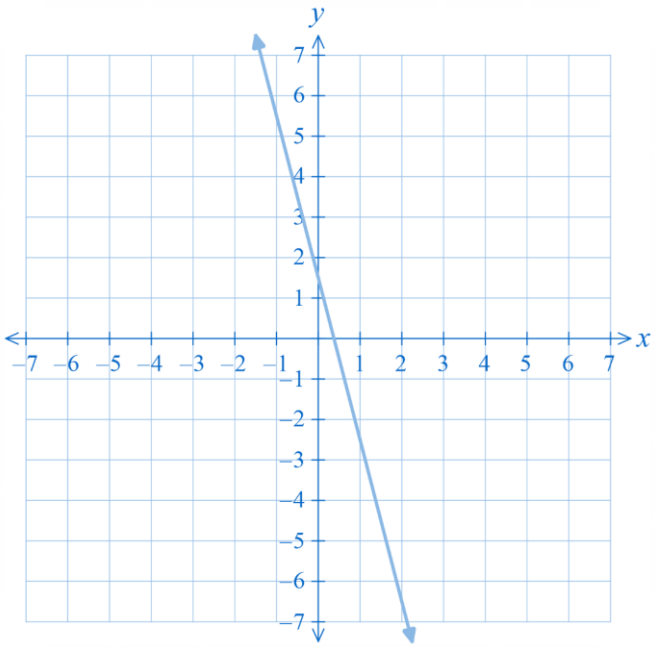
x	-2	-1	0	1	2
y					



3 Complete the table of values and graph the result.

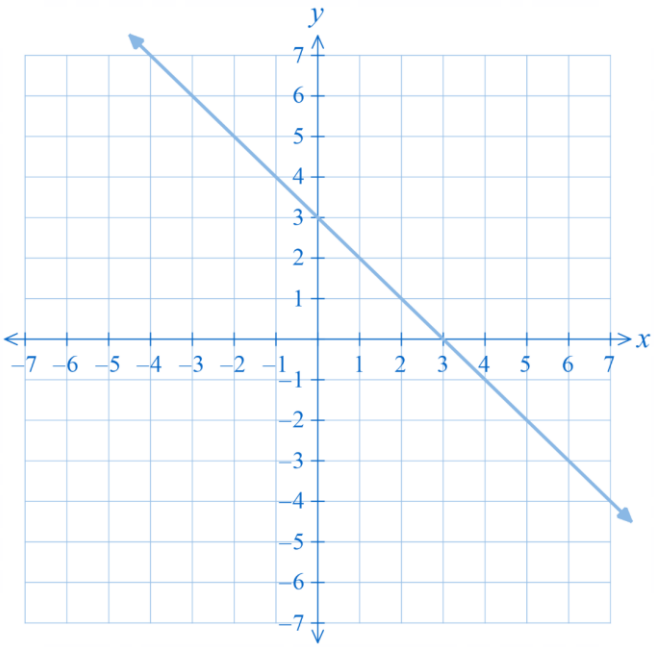
a $y = 1.5 - 4x$

x	-2	-1	0	1	2
y					



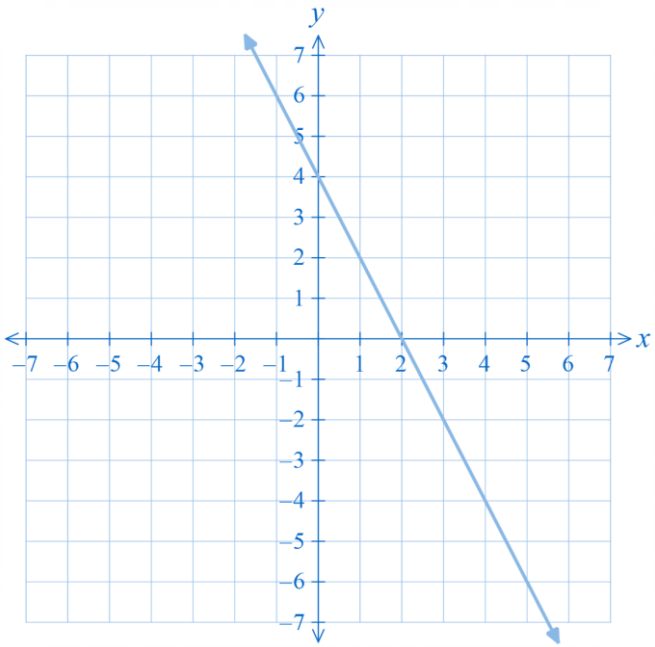
b $y = 3 - x$

x	-2	-1	0	1	2
y					



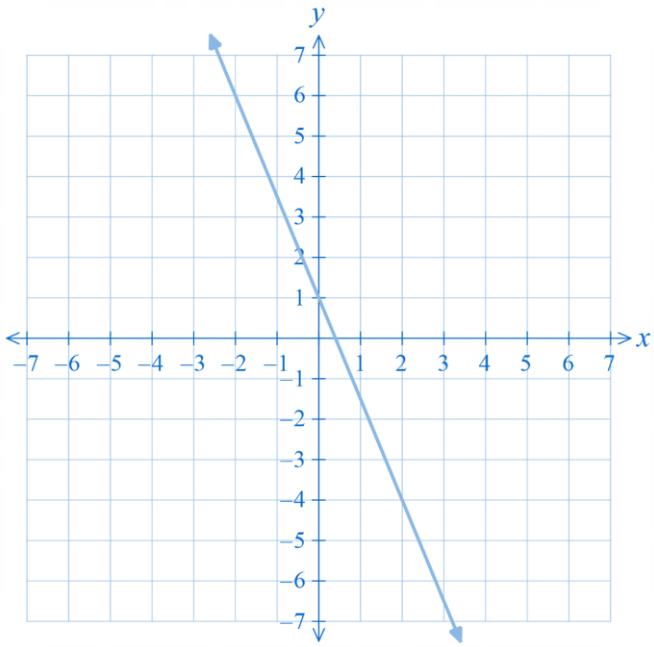
c $y = 4 - 2x$

x	-2	-1	0	1	2
y					



d $y = 1 - \frac{5}{2}x$

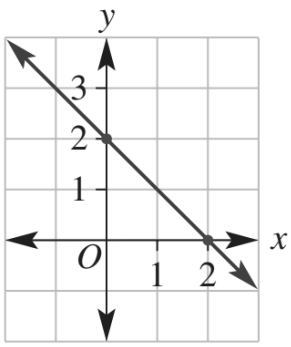
x	-2	-1	0	1	2
y					



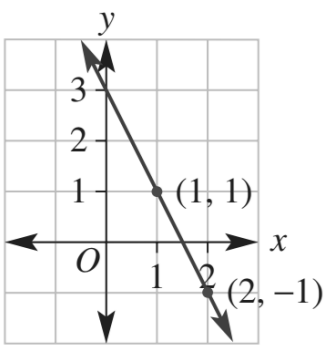
4 Match the rules A-D with the graphs a-d.

A $y = 2x + 1$ **B** $y = -x - 1$ **C** $y = -2x + 3$ **D** $y = -x + 2$

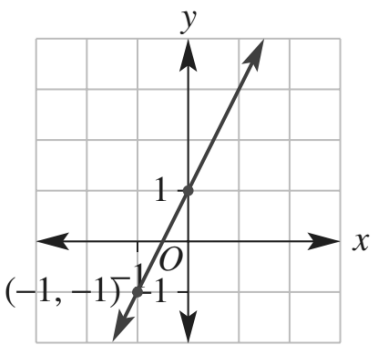
a



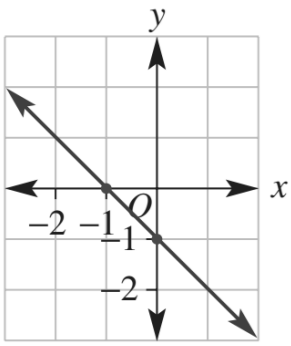
b



c



d



A – c, B – d, C – b, D – a

Horizontal and Vertical Lines

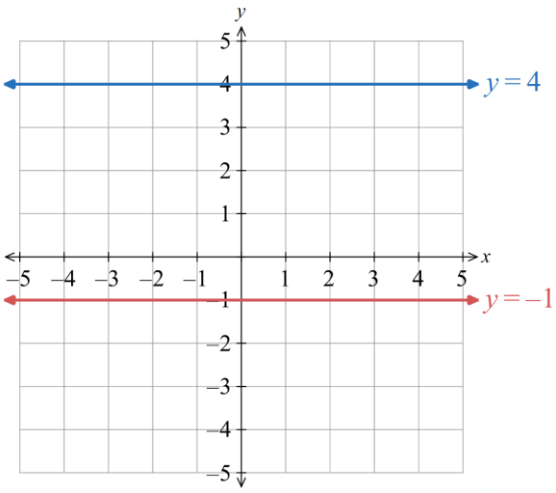
ⓘ Horizontal and Vertical Lines

$y = c$

Every y coordinate is c.

Parallel to the x axis

The x axis has equation $y = 0$



The gradient of a horizontal line is **0**:

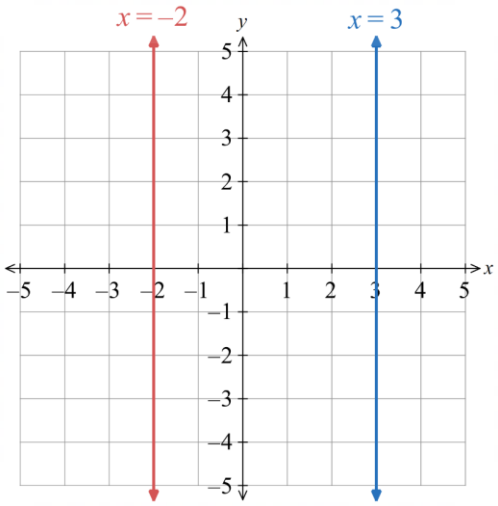
$$m = \frac{0}{\text{run}} = 0$$

$x = k$

Every x coordinate is k.

Parallel to the y axis

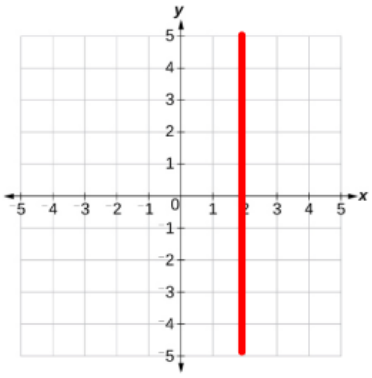
The y axis has equation $x = 0$



The gradient of a vertical line is **undefined**:

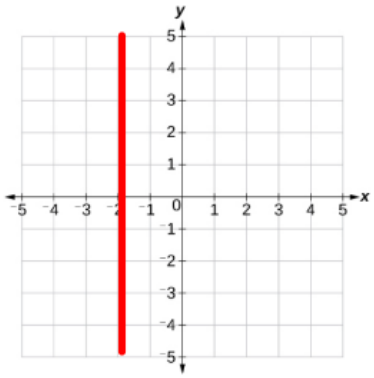
$$m = \frac{\text{rise}}{0} \quad \text{undefined}$$

Identify the equation and gradient of horizontal and vertical lines



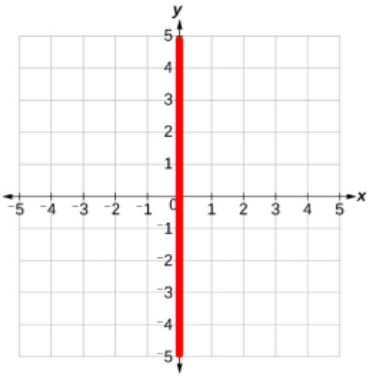
Equation: .....

Gradient: .....



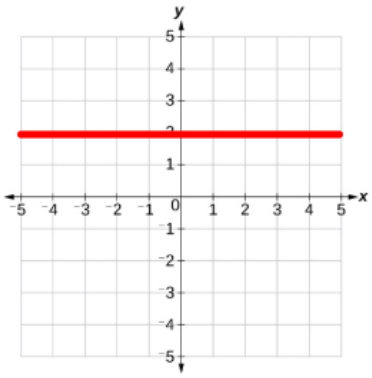
Equation: .....

Gradient: .....



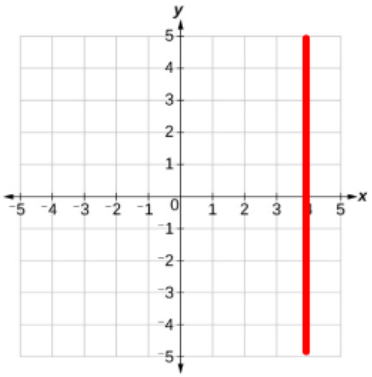
Equation: .....

Gradient: .....



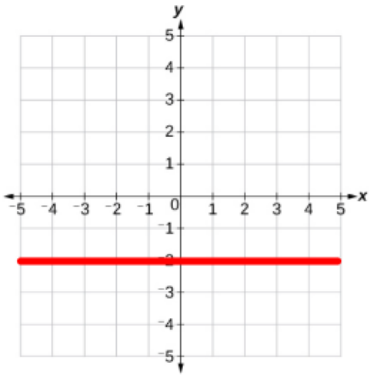
Equation: .....

Gradient: .....



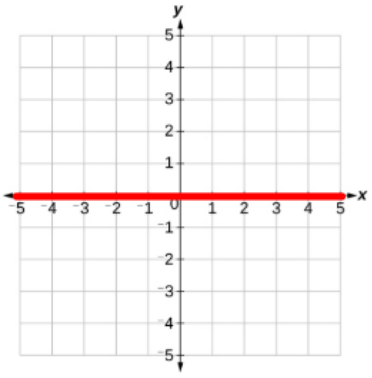
Equation: .....

Gradient: .....



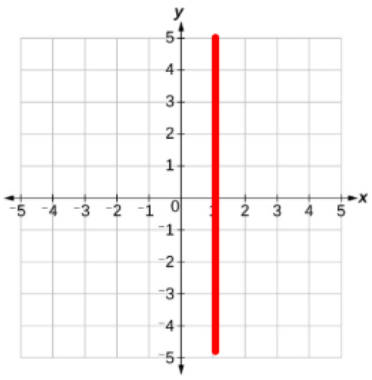
Equation: .....

Gradient: .....



Equation: .....

Gradient: .....

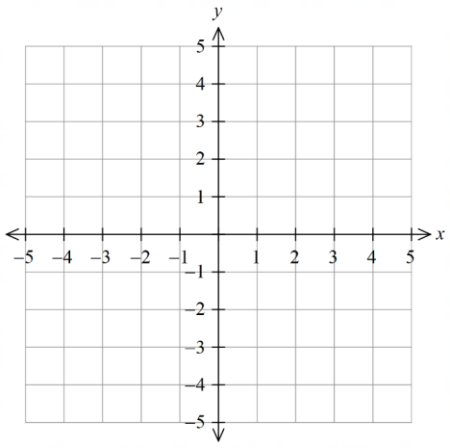


Equation: .....

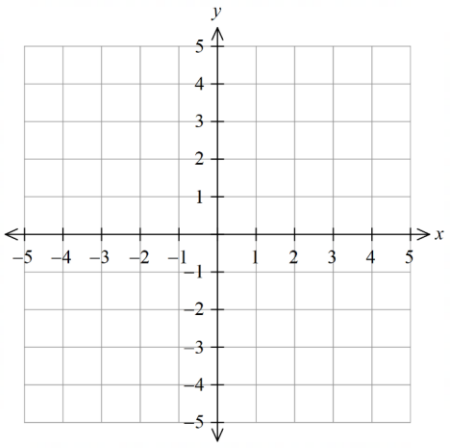
Gradient: .....

Sketch horizontal and vertical lines

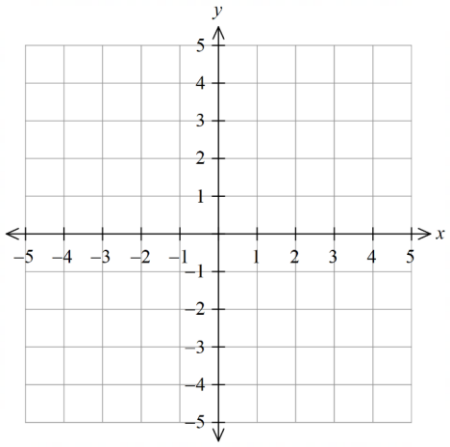
a Sketch $y = 3$



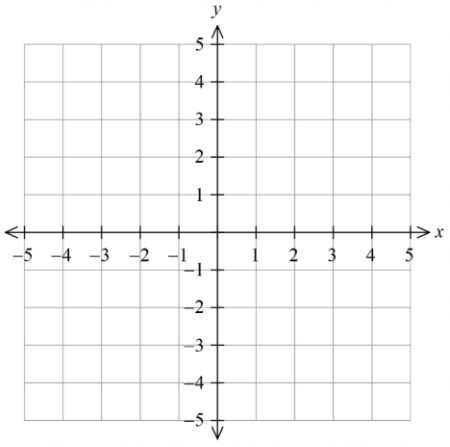
b Sketch $x = -4$



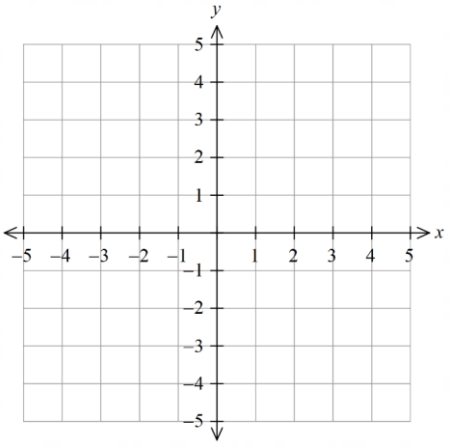
c Sketch $y = 5$



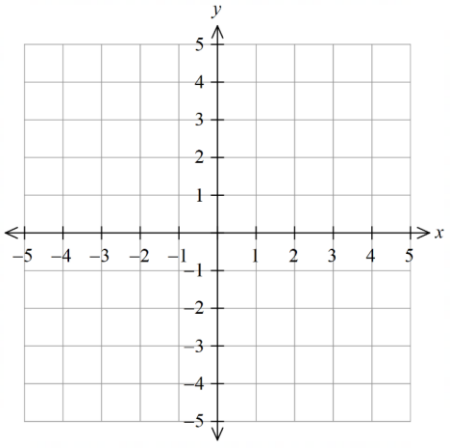
d Sketch $x = -3$



e Sketch $x = 3.5$



f Sketch $y = -2.25$

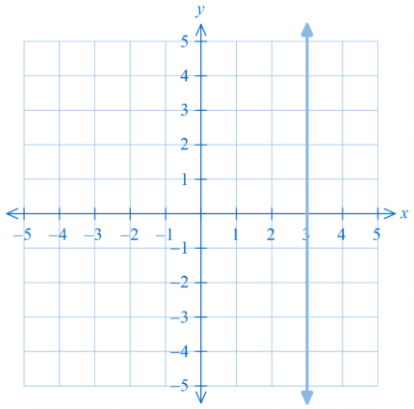


Key Ideas

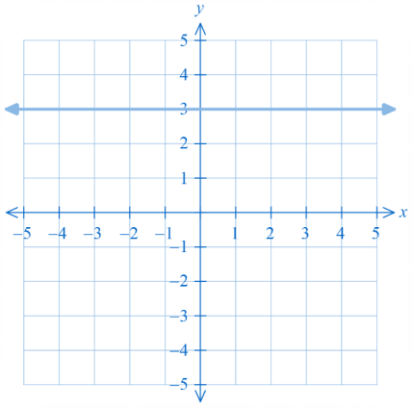
1. A …… line has an equation of the form $y = c$
2. A …… line has an equation of the form $x = k$
3. The gradient of a horizontal line is ……
4. The gradient of a vertical line is ……

1 Sketch the following lines.

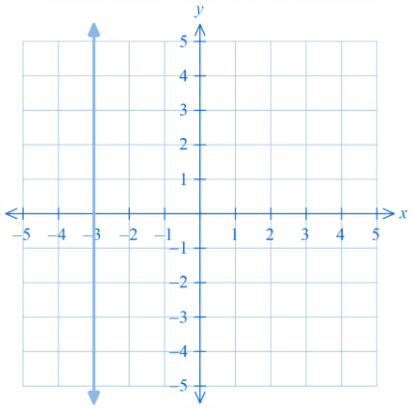
a $x = 3$



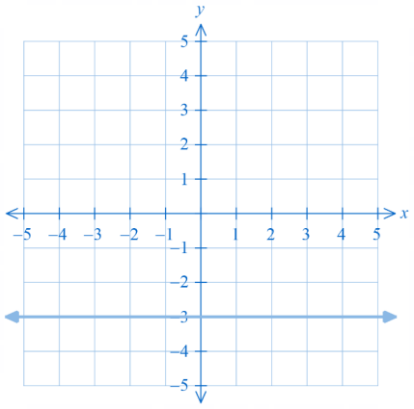
b $y = 3$



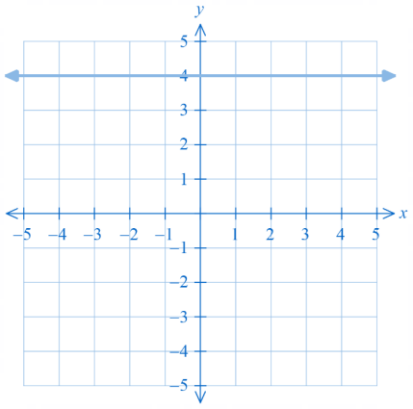
c $x = -3$



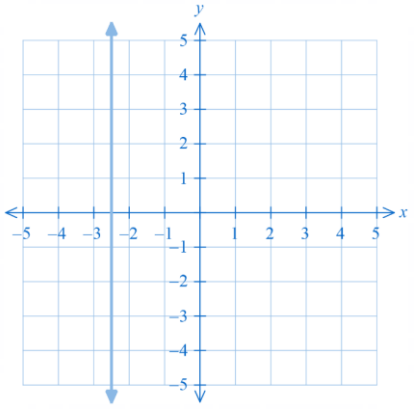
d $y = -3$



e $y = 4$

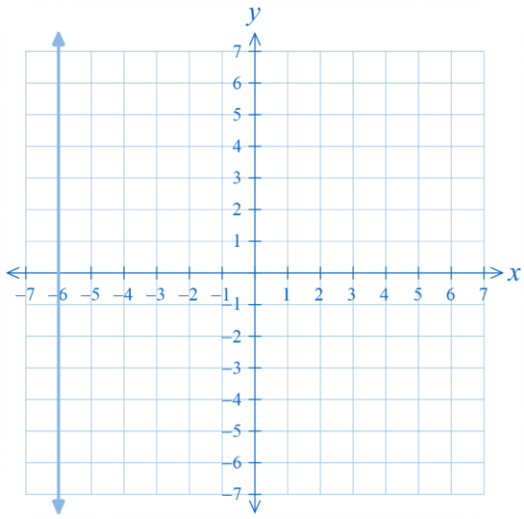


f $x = -2.5$



2 Plot the points and draw a straight line through the points.

a (-6, 2), (-6, 3), (-6, 0) and (-6,-5).



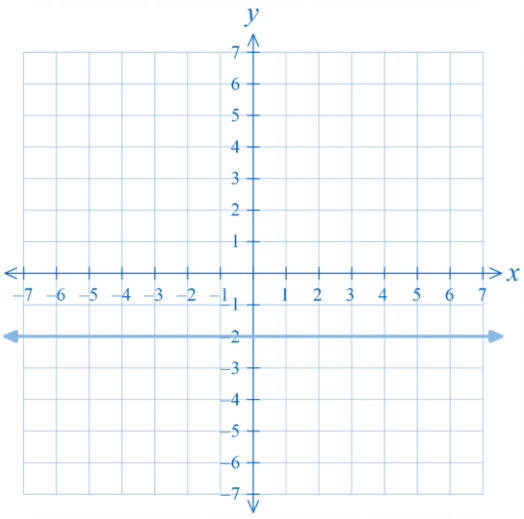
This is the line $x =$ ……

At every point on the line,

the ……-coordinate is equal to ……

–6, *x*, –6

b (-7,-2), (-5,-2) (0,-2) and (6,-2)



This is the line $y =$ ……

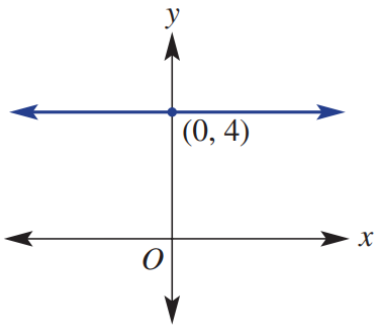
At every point on the line

the ……-coordinate is equal to ……

–2, *y*, –2

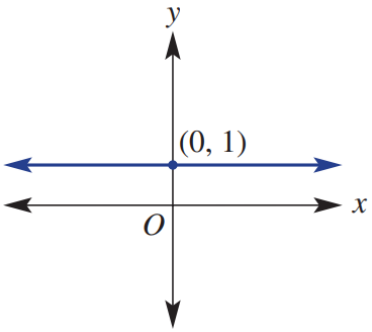
3 Write the equation of these horizontal and vertical lines.

a



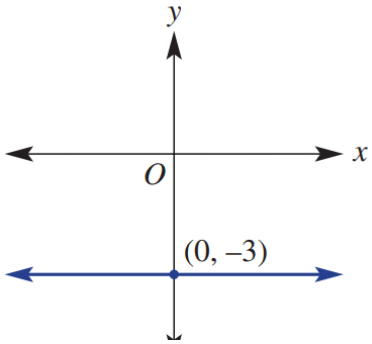
$y = 4$

b



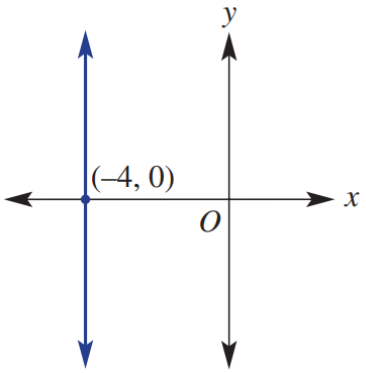
$y = 1$

c



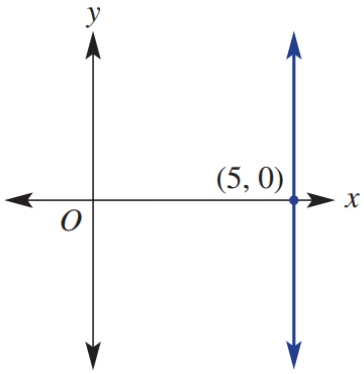
$y = -3$

d



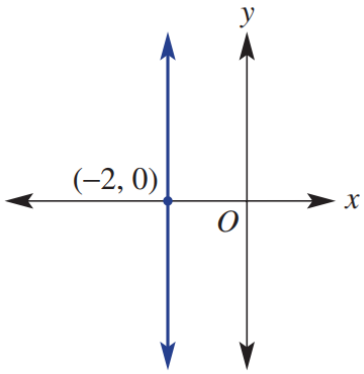
$x = -4$

e



$x = 5$

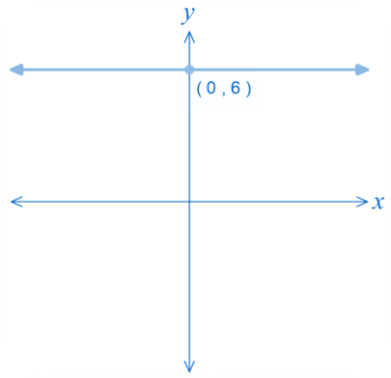
f



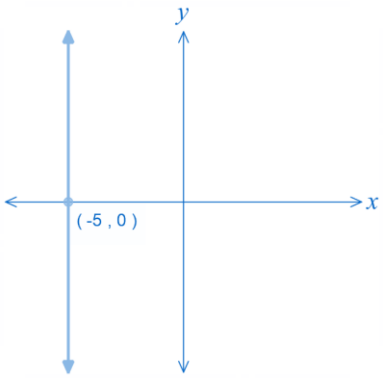
$x = -2$

4 Sketch the following lines, showing clearly the intercept.

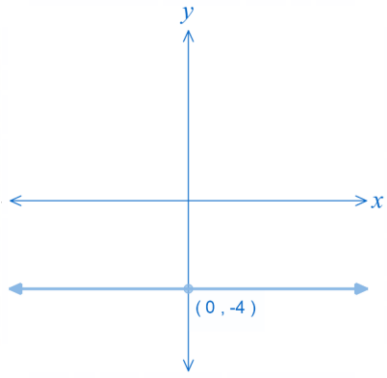
a $y = 6$



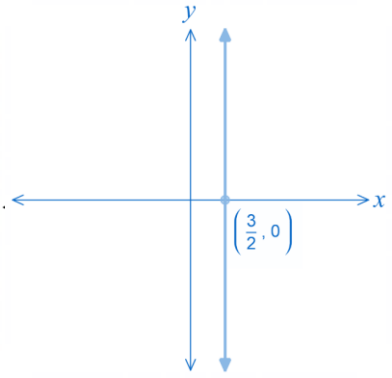
b $x = -5$



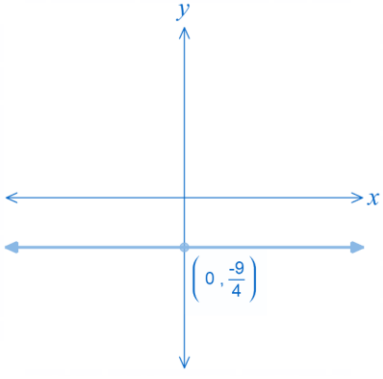
c $y = -4$



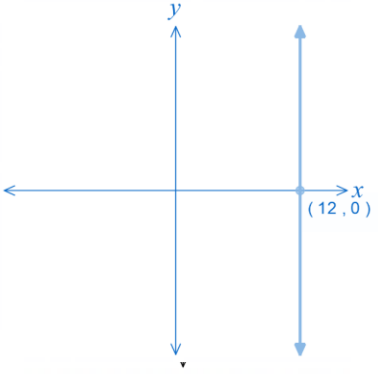
d $x = 1.5$



e $y = -\frac{9}{4}$



f $x = 12$



5 Write the equation of the line that is:

a parallel to the x-axis and passes through the point (1, 3)

$y = 3$

b parallel to the y-axis and passes through the point (5, 4)

$x = 5$

c parallel to the y-axis and passes through the point (−2, 4)

$x = -2$

d parallel to the x-axis and passes through the point (0, 0)

$y = 0$

6

a What is another name for the line $y = 0$?

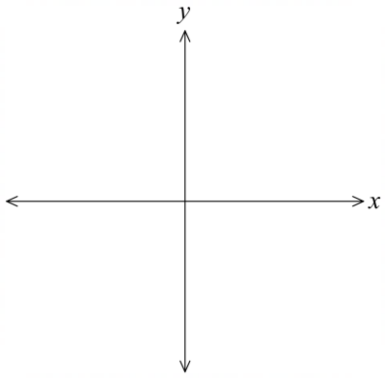
x axis

b What is another name for the line $x = 0$?

y axis

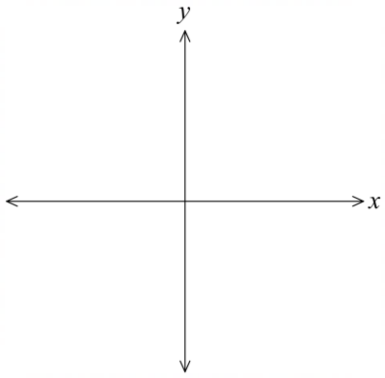
7 Find the area of the rectangle contained within the following four lines.
(Hint: sketch the 4 lines on the same axis first).

a $x = 1,\ x = -2,\ y = -3,\ y = 2$



15 u²

b $x = 0,\ x = 17,\ y = -5,\ y = -1$



68 u²

8 The lines $x = -1, x = 3$ and $y = -2$ form three sides of a rectangle.
Find the possible equation of the fourth line if:

a The area is 12 square units.

$y = 1$

b The area is 8 square units.

$y = 0$

c The perimeter is 14 units.

$y = 1$

d The perimeter is 31 units.

$y = \frac{19}{2}$

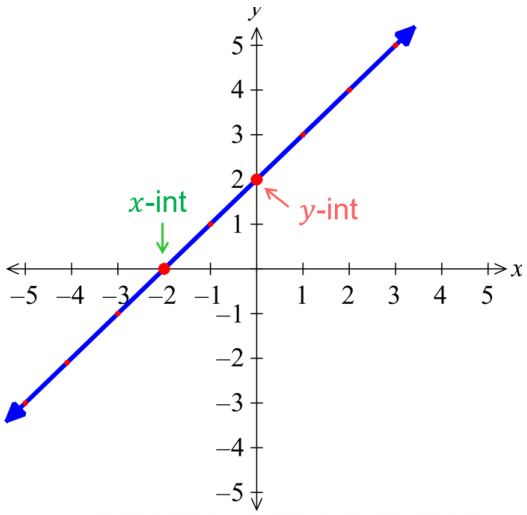
Intercepts of a Line

ⓘ x and y intercepts

x-intercept: where the line meets the x axis ($\boldsymbol{y = 0}$)

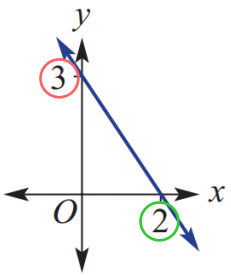
y-intercept: where the line meets the y axis ($\boldsymbol{x = 0}$)

		x-int					
$\boldsymbol{x}$	-3	-2	-1	0	1	2	3
$\boldsymbol{y}$	-1	0	1	2	3	4	5
				y-int			



ⓘ Identify x and y intercepts of a line

a



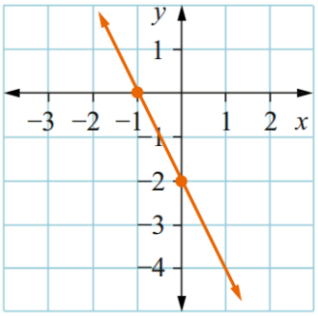
x-intercept: ..... y-intercept: .....

b

$\boldsymbol{x}$	-2	-1	0	1	2
$\boldsymbol{y}$	-6	-4	-2	0	2

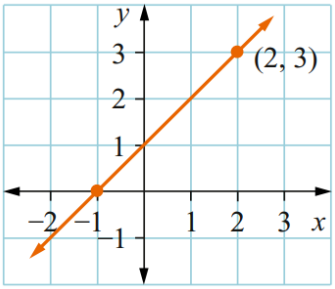
x-intercept: ..... y-intercept: .....

c



x-intercept: ..... y-intercept: .....

d



x-intercept: ..... y-intercept: .....

e

x	-2	-1	0	1	2
y	2	0	-2	-4	-6

x-intercept: ..... y-intercept: .....

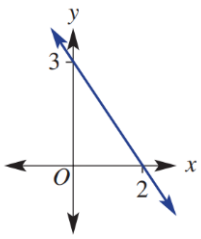
f

x	-2	-1	0	1	2
y	-9	-6	-3	0	3

x-intercept: ..... y-intercept: .....

1 Identify the intercepts from these graphs.

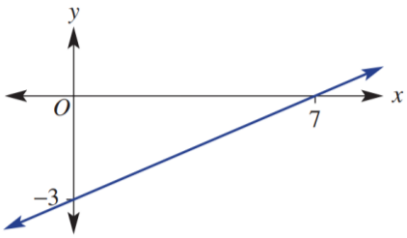
a



x-intercept: (2,0)

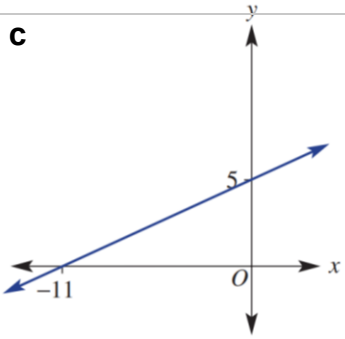
y-intercept: (0,3)

b



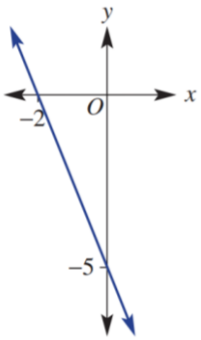
x-int: (7,0) y-int: (0,–3)

c



x-int: (–11,0) y-int: (0,5)

d



x-int: (–2,0) y-int: (0,–5)

2 Identify the x and y intercepts from these tables.

a

x	–3	–2	–1	0	1	2	3
y	4	3	2	1	0	1	2

x –int: ..... y –int: .....

x-int: (1,0) y-int: (0,1)

b

x	–3	–2	–1	0	1	2	3
y	–1	0	1	2	3	4	5

x-int: (–2,0) y-int: (0,2)

c

x	–1	0	1	2	3	4
y	10	8	6	4	2	0

x-int: (4,0) y-int: (0,8)

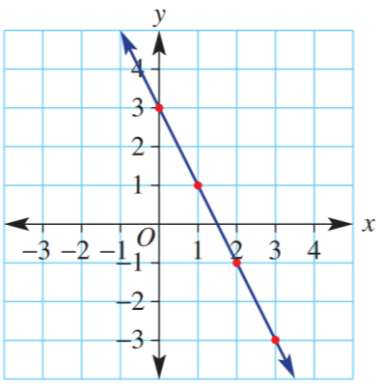
d

x	–5	–4	–3	–2	–1	0
y	0	2	4	6	8	10

x-int: (–5,0) y-int: (0,10)

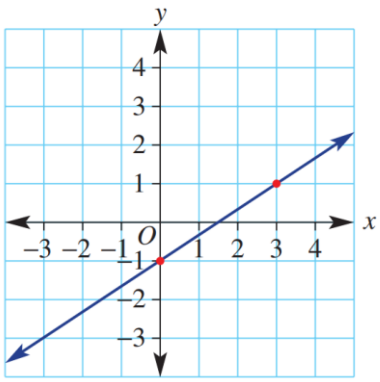
3 Identify the x and y-intercepts from these graphs.

a



x-int: (1.5,0) y-int: (0,3)

b



x-int: (1.5,0) y-int: (0,–1)

Checking if a Point Lies on a Line

Review

☐ Identify that every point on a line is a solution to the equation

a For the point $(3, 7)$ and the line $y = 2x + 1$:

What is the value of $2x + 1$ when $x = 3$?

What is the y-value of the point?

b For the point $(-1, 4)$ and the line $y = 5x - 1$:

What is the y-value of the point?

What is the value of $5x - 1$ when $x = -1$?

☐ Verify solutions to algebraic equations

a Is $x = 4$ a solution to $3x - 2 = 10$?

$LHS =$ $RHS =$

b Is $x = -3$ a solution to $2x + 5 = 1$?

$LHS =$ $RHS =$

Checking if a Point Lies on a Line

A point lies on a line when its coordinates make the line's equation true.

1. Separate the **left-hand side LHS** and **right-hand side RHS** of the equation.
2. Substitute x and y values of the point into both sides.
3. Compare: If LHS = RHS, it lies on the line. If LHS ≠ RHS, it does not lie on the line.

Example Check whether a point lies on a line by substitution

Decide whether $(2,5)$ lies on the line

$$y = 2x + 1$$

$\text{LHS} = y$
$= 5$

$\text{RHS} = 2x + 1$
$= 2(2) + 1$
$= 5$

$\text{LHS} = \text{RHS}$

$(2,5)$ lies on the line $y = 2x + 1$

Decide whether $(-2,-5)$ lies on the line

$$y = 2x + 1$$

$\text{LHS} = y$
$= -5$

$\text{RHS} = 2x + 1$
$= 2(-2) + 1$
$= -3$

$\text{LHS} \neq \text{RHS}$

$(-2,-5)$ does not lie on line $y = 2x + 1$

Guided Practice

a Decide whether $(-2,4)$ lies on the line

$$y = -x + 2$$

$\text{LHS} =$

$\text{RHS} =$

yes

b Decide whether $(11,23)$ lies on the line

$$y = 35 - x$$

$\text{LHS} =$

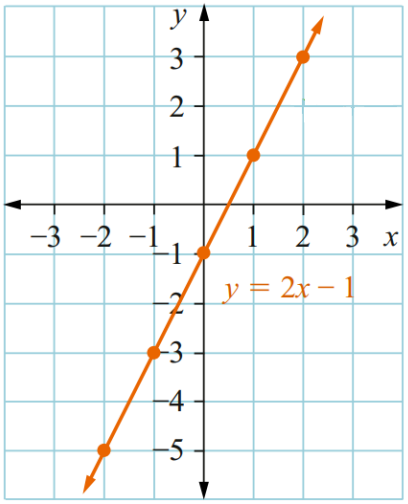
$\text{RHS} =$

no

Key Ideas

1. A point lies on a line if its coordinates make the equation ……
2. If the coordinates of a point do not satisfy the …… of the line, then it does not lie on the line.

1 Does the point $(2,2)$ lie on the line $y = 2x - 1$?



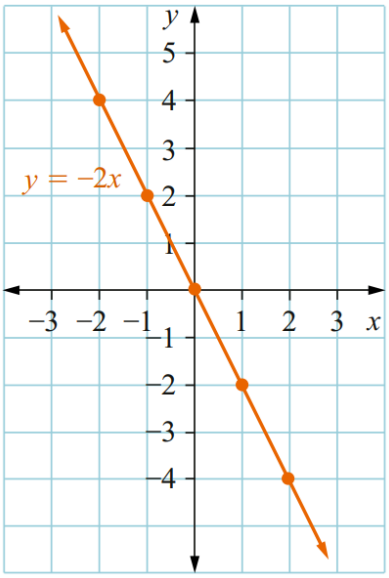
LHS $= y$	RHS $= 2x - 1$
$=$	$=$
	$=$

LHS …… RHS

The point $(2,2)$ does | doesn't lie on the line $y = 2x - 1$

no

2 Does the point $(-1, 2)$ lie on the line $y = -2x$?

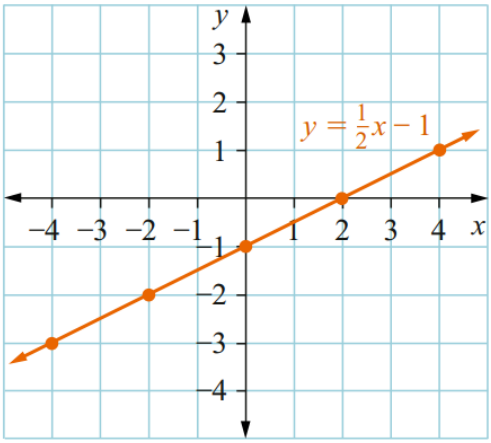


LHS $=$	RHS $=$
$=$	$=$
	$=$

LHS …… RHS

yes

3 Does the point $(-3, -2)$ lie on the line $y = \frac{1}{2}x - 1$?



LHS $=$	RHS $=$
$=$	$=$
	$=$

LHS …… RHS

no

4

a Does $(1,-3)$ lie on the line $y = x - 4$?

LHS =	RHS =
=	=

LHS …… RHS

yes

b Does $(0,2)$ lie on the line $y = 2x$?

LHS =	RHS =
=	=

LHS …… RHS

no

c Does $(4,1)$ lie on the line $y = \frac{x}{2}$?

no

d Does $(1,-2)$ lie on the line $y = -x - 3$?

no

e Does $(5,4)$ lie on the line $y = 1 - x$?

no

f Does $(-2,-1)$ lie on the line $y = 2x + 3$?

yes

g Does $(-4,-11)$ lie on the line $y = 2x - 3$?

yes

h Does $(-5,4)$ lie on the line $y = 3 - 2x$?

no

5 Mila says:

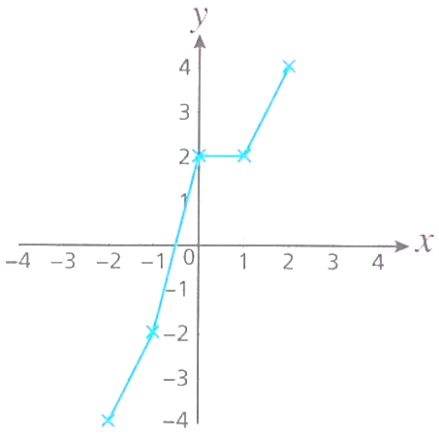


any line of the form $y = kx$ will go through the origin.

Explain why she is correct.

$x = 0, y = 0$ will always be a solution to $y = kx$

6 Rob has drawn the graph of $y = 2x$.



a Explain how you know he must have made a mistake.

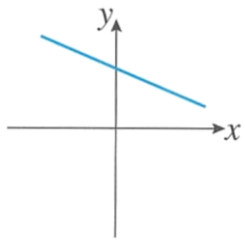
This is not a straight line.

b Which point is incorrect? Justify your answer.

(0,2), as when $x = 0, y = 0$

7 Ed has drawn the line with equation $y = -x$.

Explain how you know that Ed must have made a mistake.



This line must go through the origin as (0,0) is a solution.

8 A straight line has equation $y = 5x + 2a + 3$. The line passes through the point $(0,19)$. Work out the value of a.

Substitute into equation

$19 = 5(0) + 2a + 3$

$2a = 16$

$a = 8$

9 A line has gradient 3 and passes through the point $(1,7)$. Find its equation $y = mx + c$.

$y = 3x + c$

$7 = 3(1) + c$

$c = 4$

$\therefore y = 3x + 4$